

Social Login Implementation Guide

Flutter Application

Google & Facebook Login (Android & iOS)

1. Overview

This document describes the **complete, stable, and production-ready steps** to implement:

- Google Sign-In (Android & iOS)
- Facebook Login (Android & iOS)
- Flutter application
- Codemagic CI/CD compatible
- Facebook login **without Firebase**
- Google login **with Firebase**

This setup avoids common issues like: - White screen on iOS - OAuth popup showing project IDs - Pod / deployment target conflicts - Navigator and context crashes

2. Google Login – Android

2.1 Google Cloud Console

1. Create or select a project
2. Enable **Google Identity Services**
3. Configure **OAuth Consent Screen**
4. App name: Your actual app name
5. User support email added
6. App published (or test users added)

2.2 OAuth Client ID

- Create **Android OAuth Client ID**
- Package name:

`com.esoftsys.concessiontrackerapp`

- Add **SHA-1 & SHA-256** fingerprints

2.3 Firebase Console

1. Add Android app
2. Download `google-services.json`
3. Place file at:

```
android/app/google-services.json
```

2.4 Android Gradle Setup

android/build.gradle

```
classpath 'com.google.gms:google-services:4.4.2'
```

android/app/build.gradle

```
apply plugin: 'com.google.gms.google-services'
```

2.5 Flutter Dependencies

```
google_sign_in: ^6.2.1  
firebase_core: ^3.8.0  
firebase_auth: ^5.3.3
```

3. Google Login – iOS

3.1 Google Cloud Console

- Create **iOS OAuth Client ID**
- Note:
 - Client ID
 - Reversed Client ID

3.2 Firebase Console

1. Add iOS app
2. Bundle ID:

```
com.esoftsys.concessiontrackerapp
```

3. Download `GoogleService-Info.plist`

4. Place at:

```
ios/Runner/GoogleService-Info.plist
```

3.3 Info.plist Configuration

```
<key>GIDClientID</key>
<string>YOUR_IOS_CLIENT_ID</string>

<key>CFBundleURLTypes</key>
<array>
  <dict>
    <key>CFBundleURLSchemes</key>
    <array>
      <string>REVERSED_CLIENT_ID</string>
    </array>
  </dict>
</array>
```

3.4 iOS Deployment Target

ios/Podfile

```
platform :ios, '15.0'
```

Run:

```
cd ios
pod install
```

3.5 Fix Google Popup App Name

- Google Cloud → OAuth Consent Screen
 - Set correct **App Name**
 - Publish the app
 - Reinstall app on device
-

4. Facebook Login – Android

4.1 Facebook Developer Console

1. Create Facebook App
2. Add **Facebook Login** product
3. Add **Android platform**

4.2 Android App Configuration

- Package name:

```
com.esoftsys.concessiontrackerapp
```

- Generate **Key Hash**:

```
keytool -exportcert -alias androiddebugkey | openssl sha1 -binary |  
openssl base64
```

4.3 Android Manifest

```
<meta-data  
    android:name="com.facebook.sdk.ApplicationId"  
    android:value="@string/facebook_app_id"/>  
  
<meta-data  
    android:name="com.facebook.sdk.ClientToken"  
    android:value="@string/facebook_client_token"/>
```

4.4 strings.xml

```
<string name="facebook_app_id">YOUR_APP_ID</string>  
<string name="facebook_client_token">YOUR_CLIENT_TOKEN</string>
```

5. Facebook Login – iOS

5.1 Facebook Developer Console

- Add **iOS platform**
- Bundle ID:

```
com.esoftsys.concessiontrackerapp
```

5.2 Info.plist Configuration

```
<key>FacebookAppID</key>
<string>YOUR_APP_ID</string>

<key>FacebookClientToken</key>
<string>YOUR_CLIENT_TOKEN</string>

<key>FacebookDisplayName</key>
<string>Your App Name</string>

<key>CFBundleURLTypes</key>
<array>
  <dict>
    <key>CFBundleURLSchemes</key>
    <array>
      <string>fbYOUR_APP_ID</string>
    </array>
  </dict>
</array>

<key>LSApplicationQueriesSchemes</key>
<array>
  <string>fbapi</string>
  <string>fb-messenger-share-api</string>
  <string>fbauth2</string>
  <string>fbshareextension</string>
</array>
```

6. Flutter – Facebook Auth Service

Dependency

```
flutter_facebook_auth: ^7.0.0
```

Service Implementation

```
class FacebookAuthService {
  Future<Map<String, dynamic>?> signInWithFacebook() async {
```

```

    final result = await FacebookAuth.instance.login(
      permissions: ['email', 'public_profile'],
    );

    if (result.status == LoginStatus.success) {
      final userData = await FacebookAuth.instance.getUserData(
        fields: "name,email,picture.width(200)",
      );
      return userData;
    }
    return null;
  }
}

```

7. Navigation Safety

Always guard context usage:

```
if (!mounted) return;
```

Use `pushReplacement` after login.

8. Codemagic Notes

Android

- No special config required
- Ensure SHA keys added

iOS

- Deployment target ≥ 15.0
- Upload `GoogleService-Info.plist`
- Automatic signing recommended

9. Final Checklist

✓ Google Login Android
 ✓ Google Login iOS
 ✓ Facebook Login Android
 ✓ Facebook Login iOS
 ✓ No Firebase for Facebook
 ✓ Codemagic compatible
 ✓ No white screen issues

10. Next Enhancements (Optional)

- Apple Sign-In (App Store requirement)
- Backend token validation
- Secure session storage
- Refresh token handling

End of Document